

Project Initialization and Planning Phase

Date	27 June 2026
Team ID	XXXXXX
Project Title	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed **Credit Card Approval Prediction System** aims to simplify the credit card approval process using Machine Learning technology. The system predicts whether a credit card application is likely to be approved or rejected based on the applicant's personal and financial information.

Project Overview	
Objective	The primary objective is to develop a Machine Learning-based Credit Card Approval Prediction System that provides fast and accurate approval predictions through a simple web application.
Scope	The project focuses on collecting applicant information, preprocessing the data, applying a trained Machine Learning model, and displaying instant credit card approval predictions using a Flask-based web application.
Problem Statement	
Description	Traditional credit card approval involves manual verification, making the process time-consuming and sometimes inconsistent. Applicants often do not know whether they are eligible before applying.

Impact	Delayed approval decisions reduce customer satisfaction, increase the workload of financial institutions, and may lead to inconsistent preliminary assessments.
Proposed Solution	
Approach	Develop a Machine Learning model using Scikit-learn, integrate it with a Flask backend, preprocess applicant information, and generate credit card approval predictions through a responsive web application.
Key Features	• Machine Learning-based credit card approval prediction. • Real-time prediction using Flask. • User-friendly web interface. • Displays approval/rejection result with confidence score.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU specifications	Intel Core i5 or above
Memory	RAM specifications	8 GB RAM
Storage	Space for dataset, models and project files.	10 GB Free Disk Space
Software		
Frameworks	Backend Framework	Flask

Libraries	Machine Learning Libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Joblib
Development Environment	IDE	Visual Studio Code
Data		
Data	Dataset	Kaggle Credit Card Approval Prediction Dataset (application_record.csv & credit_record.csv)